LIVE — MAINNET RUNNING & MINING ACTIVE

TARCOIN

[TAR]

A UTXO-Based Decentralized Cryptocurrency Ecosystem
Built on Battle-Tested Proof-of-Work Architecture

VERSION	PUBLISHED	ALGORITHM	LICENSE
1.0.0	June 2026	SHA256d PoW	MIT

50B	50,000	10 min	400K	P2P:19333
TOTAL SUPPLY (TAR)	BLOCK REWARD (TAR)	BLOCK TIME	HALVING INTERVAL	MAINNET PORT

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ABSTRACT

TARCOIN (TAR) is a fully decentralized, UTXO-based cryptocurrency built on a battle-tested Proof-of-Work blockchain foundation derived from Bitcoin Core. It inherits the complete security model — SHA256d Proof-of-Work, UTXO transaction model, secp256k1 elliptic curve cryptography, ECDSA signatures, Bitcoin Script, Merkle trees, difficulty retargeting, SegWit, and longest-chain consensus — while introducing a distinct identity, custom supply parameters, and a complete web/mobile ecosystem.

Unlike experimental blockchain projects that modify consensus rules or introduce centralized components, TARCOIN remains faithful to the proven principles of decentralized consensus, ensuring maximum security, ASIC miner compatibility, and long-term stability.

SECTION 01

Introduction

1.1 Philosophy

TARCOIN follows the original Bitcoin philosophy: a peer-to-peer electronic cash system that is decentralized, permissionless, and immutable. Every design decision prioritizes **security over convenience**, **decentralization over speed**, and **immutability over flexibility**.

1.2 Design Goals

1. **Battle-Tested Security** — Preserve all Bitcoin consensus rules and cryptographic primitives
2. **ASIC Mining Compatibility** — SHA256d algorithm supports existing Bitcoin mining hardware
3. **Immutable Supply** — Fixed 50 billion TAR supply, enforced by consensus
4. **Decentralized Operation** — No admin controls, no backdoors, no central authority
5. **Privacy-Preserving Development** — Secure, anonymous development and deployment
6. **Production Infrastructure** — Explorer, API, mining pool, wallets, monitoring

SECTION 02 **Technical Architecture**

2.1 Blockchain Foundation

TARCOIN is derived from Bitcoin Core v31.x. Only the following components are modified:

- **Supply constants** (`amount.h`)
- **Network parameters** (`chainparams.cpp` — magic bytes, ports, DNS seeds, checkpoints)
- **Address prefixes** (bech32: `tar1`, base58: `T`)
- **Branding** (coin name, ticker, UI identity)
- **Genesis block** (unique hash, treasury allocation)

Everything else — validation logic, consensus engine, networking, mempool, wallet encryption, RPC interface — remains **identical to upstream Bitcoin Core**, ensuring full compatibility with the Bitcoin security model.

2.2 Cryptographic Primitives

COMPONENT	STANDARD
Hashing Algorithm	SHA256d (double SHA-256)
Elliptic Curve	secp256k1
Signature Algorithm	ECDSA
Address Format	Bech32 (SegWit) and Base58 (Legacy)
Merkle Trees	Binary Merkle (Bitcoin standard)
Script Language	Bitcoin Script

2.3 Supply Schedule

TOTAL SUPPLY
50,000,000,000 TAR

ECOSYSTEM TREASURY (20%)
10,000,000,000 TAR

PUBLIC MINING SUPPLY (80%)
40,000,000,000 TAR

BLOCK REWARD (ERA 1)
50,000 TAR

Emission Schedule

```
Era 1:  Block       0 – 400,000  │  50,000 TAR/block  │  ~7.6 yrs  │  20,000,000,000 TAR
Era 2:  Block 400,001 – 800,000  │  25,000 TAR/block  │  ~7.6 yrs  │  10,000,000,000 TAR
Era 3:  Block 800,001 – 1,200,000│  12,500 TAR/block  │  ~7.6 yrs  │   5,000,000,000 TAR
Era 4:  Block 1,200,001–1,600,000│   6,250 TAR/block  │  ~7.6 yrs  │   2,500,000,000 TAR
...emission continues halving every 400,000 blocks until max supply is reached
```

SECTION 03

Network Architecture

3.1 Network Parameters

PARAMETER	MAINNET	TESTNET	REGTEST
P2P Port	19333	29333	18444
RPC Port	19332	29332	18443
Magic Bytes	0xfabfb5da	0x0b110907	0xfabfb5da
Address Prefix	tar1 / T	tar1 / T	tar1 / T

3.2 Mainnet Parameters

PARAMETER	VALUE
Coin Name	TARCOIN
Symbol	TAR
Consensus	SHA256d Proof-of-Work
Block Time	10 minutes (Bitcoin-standard)
Initial Block Reward	50,000 TAR
Halving Interval	400,000 blocks
Maximum Supply	50,000,000,000 TAR
Mineable Supply	40,000,000,000 TAR (80%)
Ecosystem Treasury	10,000,000,000 TAR (20%)
P2P Port	19333
RPC Port	19332
Bech32 Address Prefix	tar1

3.3 DNS Seed Nodes

Seed nodes are deployed across multiple cloud providers (DigitalOcean, AWS, Hetzner, Vultr) to ensure network resilience. Each seed node runs a full archive node.

3.4 Peer-to-Peer Protocol

TARCOIN uses the standard Bitcoin-compatible P2P protocol:

- `Version/VerAck` — initial handshake
- `Inv/GetData` — inventory propagation
- `Headers/GetHeaders` — block header sync
- `Block/Tx` — data propagation
- `Addr` — peer discovery
- `Ping/Pong` — keepalive
- `FeeFilter` — fee relay

SECTION 04

Consensus Rules

4.1 Proof-of-Work

SHA256d Proof-of-Work with difficulty retargeting every 2016 blocks (approximately 2 weeks). Difficulty adjusts based on the time taken to mine the previous 2016 blocks, maintaining a ~10-minute average block time.

4.2 Block Validation

Block validation follows the Bitcoin Core consensus standard:

- Verify block header hash meets difficulty target
- Verify Merkle root matches transactions
- Verify all transaction inputs reference valid UTXOs
- Verify no double-spends within the block
- Verify block size ≤ 1MB (base) / 4MB (SegWit)
- Verify signature operations limits
- Verify coinbase maturity

4.3 Genesis Block

The TARCOIN genesis block was successfully mined and verified on mainnet.

Block Hash	000074c6359f78730790275ea21bbd53f0bc3249604470bad49b9753f44bd7e0
Merkle Root	1fa777a38f96e44bb26591573ed2b22d5b40d7a63067201a40ad3b214152b749
Timestamp	1748304000 (2025-05-27 00:00:00 UTC)
nBits	0x1f00ffff
Nonce	15878
Block Reward	50,000 TAR
PoW Target	0000ffff00000000000000000000000000000000000000000000000000000000
powLimit	0000ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff
Height	0

The genesis block is **hardcoded** into `chainparams.cpp` and verified by `assert()` on every node startup.

SECTION 05 **Ecosystem Treasury Security**

5.1 Offline Generation

The ecosystem treasury wallet is created entirely offline using an **air-gapped machine** that has never been connected to the internet.

1. Generate wallet on air-gapped machine
2. Record public address only (publicly verifiable)
3. Create multiple encrypted backups (USB, paper, steel)
4. Transfer public address to development environment
5. Hardcode address into `chainparams.cpp`
6. Destroy intermediate systems

5.2 Private Key Security

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🚫 **MUST NEVER** touch any internet-connected system

🚫 **MUST NEVER** be stored in repositories or cloud systems

🚫 **MUST NEVER** be logged or screenshotted

✅ **MUST** have geographically separated backups

✅ **MUST** follow industry-standard cold storage best practices

5.3 Transparency & Immutability

- The ecosystem treasury allocation is permanent and immutable
- No mechanism exists to modify the supply or allocation
- All treasury transactions are publicly recorded on the blockchain
- No individual or entity has unilateral control over treasury funds
- The blockchain operates independently of any developer or entity

SECTION 06

Security Model

6.1 Attack Mitigation

51% Attack Protection (Early Network)

- Encourage distributed mining participation
- Deploy multiple mining pools
- Public mining awareness campaign
- Optional temporary checkpoints if required (for initial security only)
- Rapid hashrate growth through ASIC compatibility

Reorg Protection

- Bitcoin-standard reorganization handling
- 100-block maturity for coinbase outputs
- Merkle tree verification
- Checkpoints at regular intervals

Double-Spend Protection

- Full node validation
- Mempool orphan detection
- Replace-by-fee (RBF) support
- UTXO set verification

6.2 Wallet Security

- AES-256 encrypted wallet.dat files
- BIP39/BIP32 mnemonic seed phrases
- Passphrase protection
- Offline transaction signing (cold storage)
- Hardware wallet compatibility roadmap

6.3 Network Security

- Full node validation only (no SPV reliance)
- Tor/I2P support for privacy
- Encrypted P2P connections (BIP324)

- DoS protection (limit counters, penalty system)

SECTION 07 **Infrastructure Architecture**

7.1 Explorer

The blockchain explorer provides real-time block, transaction, and address lookup:

- 📦 Real-time block lookup
- 💸 Transaction explorer
- 📊 Difficulty & hashrate charts
- 🏦 Supply tracking & rich list
- 🔴 WebSocket live updates
- ⚡ Redis-cached API

7.2 REST API

RESTful API with comprehensive blockchain data endpoints, Swagger documentation, rate limiting, and DDoS protection.

7.3 Mining Pool

Full-featured mining pool on Stratum port 3333 with worker management, share tracking via Redis, payout engine, and ASIC miner compatibility.

7.4 Website

Built with Next.js featuring a cyberpunk/gold-black aesthetic, Framer Motion animations, Three.js particle effects, responsive design, and SEO optimization.

SECTION 08

Deployment Architecture

8.1 Docker Stack

All services are containerized for deterministic deployment.

SERVICE	DESCRIPTION
tarcoind	Blockchain full node
explorer	Blockchain explorer frontend + backend
api	REST API server
mining-pool	Mining pool + Stratum server
website	Next.js public website
nginx	Reverse proxy (SSL termination, DDoS protection)
redis	Cache layer
postgres	Relational database
prometheus / grafana / loki	Monitoring & observability stack

8.2 Infrastructure Requirements

SERVICE	CPU	RAM	STORAGE	NETWORK
tarcoind (archive)	8 cores	16 GB	1+ TB SSD	1 Gbps
tarcoind (pruned)	4 cores	8 GB	50 GB SSD	100 Mbps
Explorer + API	4 cores	8 GB	100 GB SSD	1 Gbps
Mining Pool	4 cores	8 GB	50 GB SSD	1 Gbps
Website	2 cores	4 GB	20 GB SSD	100 Mbps
Monitoring	2 cores	4 GB	100 GB SSD	100 Mbps

8.3 Cloud Providers Supported

- DigitalOcean
- AWS (EC2, EKS)
- Hetzner
- Vultr

SECTION 09

Testing & Audit

9.1 Mandatory Audits

#	AUDIT TYPE	SCOPE
1	Consensus Audit	Verify all consensus rules match Bitcoin Core
2	Supply Audit	Verify supply constants, halving schedule, treasury allocation
3	Wallet Penetration Test	Secure key storage, encryption, signing
4	RPC Security Audit	No unauthorized access, injection, or exposure
5	API Security Testing	Rate limiting, SQL injection, XSS, CSRF
6	Memory Analysis	No leaks in long-running processes
7	Reorg Attack Simulation	Network partition handling
8	51% Attack Simulation	Chain reorganization resilience
9	Mining Stress Test	Pool and Stratum server capacity
10	P2P Network Test	Peer discovery, propagation, DoS resilience

9.2 Deterministic Builds

All releases must be:

- Deterministic (same source → same binary)
- Reproducible (anyone can verify)
- Publicly auditable
- Signed with release keys

SECTION 10 Launch Checklist

PRE-LAUNCH

18 / 18 COMPLETE

- ✓ Genesis block mined and verified (`000074c6...f44bd7e0`)
- ✓ TARCOIN Core v31.x compiled and running
- ✓ Mainnet node boots and passes PoW validation
- ✓ `powLimit` aligned with genesis `nBits` (`0x1f00ffff`)
- ✓ Genesis hash and merkle root `assert()` verified on startup
- ✓ `CLIENT_VERSION_IS_RELEASE = true` set
- ✓ Seed nodes deployed (DNS seeds live)
- ✓ Explorer backend verified
- ✓ Wallet binaries built and signed
- ✓ Mining pool configured and tested
- ✓ Infrastructure stress-tested
- ✓ Monitoring stack operational
- ✓ DNS records configured
- ✓ SSL certificates issued
- ✓ Cloudflare configured
- ✓ CI/CD pipeline validated
- ✓ Deterministic builds confirmed
- ✓ Source code publicly released on GitHub

LAUNCH

8 / 8 COMPLETE

- ✓ Seed nodes go online

- ✓ Explorer goes online
- ✓ Mining pool activates
- ✓ Wallet downloads available
- ✓ Website published
- ✓ API operational
- ✓ Social media announcements
- ✓ Mining campaigns begin

POST-LAUNCH

4 / 4 COMPLETE

- ✓ Network monitoring active
- ✓ Community support channels live
- ✓ Bug bounty program launched
- ✓ Ongoing security audits in progress

Transparency Statement: All critical network parameters, supply allocations, genesis data, and consensus rules are publicly documented and independently verifiable from the TARCOIN source code.

SECTION 11 Governance

TARCOIN operates without any centralized governance authority. The blockchain is governed solely by its immutable consensus rules, enforced by network participants running full nodes.

- 🔒 No foundation, corporation, or legal entity controls the network
- 🔒 No administrative keys or backdoors exist
- 🔒 No single entity has authority to upgrade or modify the protocol
- 🔒 No governance tokens or on-chain voting mechanisms
- 🌐 Development is community-driven through open-source contributions
- 🔒 No individual can alter supply schedule, consensus rules, or treasury after genesis

The ecosystem treasury of 10 billion TAR is allocated at genesis to a publicly verifiable cold storage address. The treasury allocation is permanently recorded in the genesis block and remains publicly auditable on-chain.

SECTION 12 Conclusion

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TARCOIN represents a return to the foundational principles of decentralized cryptocurrency — a secure, permissionless, UTXO-based blockchain network that anyone can mine, use, or build upon. By preserving a battle-tested Proof-of-Work codebase while introducing a unique identity, custom supply parameters, and a complete ecosystem, TARCOIN provides a reliable foundation for long-term value storage and peer-to-peer transactions.

TARCOIN is not an experiment.

It is battle-tested Proof-of-Work architecture, reborn for a new era.

References

1. Nakamoto, S. "Bitcoin: A Peer-to-Peer Electronic Cash System" (2008)
2. Bitcoin Core — https://github.com/bitcoin/bitcoin
3. BIP 32 — Hierarchical Deterministic Wallets
4. BIP 39 — Mnemonic Code for Generating Deterministic Keys
5. BIP 141 — Segregated Witness (SegWit)
6. BIP 173 — Bech32 Address Format